

Leandro Laenen

Unreal Engine Technical Artist focused on optimization, shaders, Niagara systems and development pipelines & tools.

✉ Leandro@gradientgames.net 📍 Genk, Belgium 🌐 [My Projects](#)

Skills

→ Graphics Programming and VFX

Materials, Niagara, HLSL, Post-Process Shaders

→ Programming and Scripting

Blueprints, C++

→ Pipeline and Tools

UE5 PCG, Blender Geometry Nodes, Source Control (Perforce, Git)

→ Optimization

Unreal Insights, Performance Profiling

→ Technical Art Generalist

Hard-Surface 3D Modeling, PBR Texturing

Experience

Gradient Games

Genk, Belgium

Technical Artist / Sole Developer

September 2023 - February 2026

Programmed, designed, and shipped two commercial titles on Steam, driving 20k+ organic wishlists, managing the full lifecycle from pre-production to release and post-launch patching.

<https://gradientgames.net/>

Projects

Midnight Mayhem

2024 - 2025

Technical Artist / Developer

- **Solved a deferred rendering pipeline limitation** for shadow-based mechanics by decoupling character visibility from the main shadow pass, engineering a localized 64x64 depth render target system costing only ~0.024ms per light source.
- **Profiled and optimized RAM, VRAM and GPU usage** (draw calls, material complexity, texture budgets) and created Niagara GPU systems to render micro-fauna at near-zero performance cost.
- **Targeted and maintained a 50fps framerate target** to secure official Steam Deck Verification.

[Midnight Mayhem Technical Breakdown Available Online](#)

The Last Explorer

2023 - 2024

Technical Artist / Developer

- **Created PCG tools** to accelerate open world development and iteration.
- **Developed a Gerstner wave water shader** and buoyancy system, blending visual fidelity with real-time technical art constraints.

Technical Breakdowns

• The Shadow-Driven Opacity Paradox

Created a character that is invisible in the shadows by creating a custom shadow system to bypass deferred rendering limitations.

<https://leandrolaenen.com/projects/Wdage3>

• Nordic Island Creation Pipeline

Engineered a pipeline to quickly create nordic islands using Blender Geometry Nodes, a UE5 complex auto-material and UE5 PCG.

<https://leandrolaenen.com/projects/1L1ZXq>

• 100K GPU Frogs

Created ambient wildlife swarm systems capable of handling tens of thousands of rats and frogs at extremely low cost on a mid-range GPU.

<https://leandrolaenen.com/projects/EYqXO2>

• Flat-Surface 3D Sinkhole With Physics Simulation

Engineered a way of rendering a 3d sinkhole as a flat surface without render layers, stencils or post processing. The system also allows for physics such as players falling into the hole.

<https://leandrolaenen.com/projects/mOZ3re>

Awards

Nominated: Multiplayer Game of the Year

Belgian Game Awards

[Midnight Mayhem Awards Page](#)

Education

Independent Study

- Self-directed curriculum focusing on real-time graphics, HLSL, C++, and procedural pipelines.
- Developed and shipped two commercial titles concurrently with technical training.

Languages

English (Fluent)

Dutch (Native)

French (Advanced Listening & Reading / Elementary Speaking)

Interests

Scuba Diving

About

Driven by a lifelong fascination with games, real-time graphics and 3D pipelines. I began experimenting with Blender at age 10 before discovering Unreal Engine a few years later, pivoting early on into the intersection of art and programming.

Having published two commercial titles, I gained valuable insight into the entire development lifecycle. This experience solidified my true passion: stepping away from general operations to focus on solving complex technical art challenges, optimizing performance, and building robust pipelines within a dedicated team.